USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 24 LAB - Null Steering
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.9: I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

This packet is the L24 Lab, in three parts, and you turn it in once, complete, before Lesson 25. Do Part 1 on your own before the lab period: the predictions you will test in class. Part 2 is your own measurements, from your own kit, recorded in class. Do Part 3 on your own after the lab: compare the measurements with the predictions and explain the differences.

Grading: one submission, counted toward your engagement grade and graded as a complete, documented attempt.

The array is the PHASER's: $N = 8$ elements, E1 to E8, on a $d = 14$ mm pitch. Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. The Element Gains sliders are voltage amplitudes in percent of full scale, linear in voltage. Use $c = 3.00 \times 10^8$ m/s, and report every level relative to the frozen uniform trace's peak unless a question asks otherwise.

Part 1: Pre-Lab (individual, before the lab period)

Every question here produces a prediction you check in Part 2, at the HB100 frequency your kit reported with **Find HB100** in an earlier lab. If you did not record one, use 10.525 GHz, the HB100's nominal frequency, and say so. Lesson 24 works its examples at 10.0 GHz, which is no kit's frequency, so its numbers show the method but are not your answers.

1. **Your kit's notch** (Procedure A, Table A). Hold the beam at broadside and null an interferer at $\theta_1 = +22.5^\circ$.

   (a) Find kd and the element-to-element phase $\psi_1 = kd \sin\theta_1$ at the null angle.

   f = _____

   ψ_1 = _____

   (b) Write a short Python script (NumPy) that forms $w_n = 1 - r_n\, e^{-jn\psi_1}$, $n = 0, \ldots, 7$, with $r_n = \frac{1}{8}\sum_n e^{jn\psi_1}$, and converts the weights to the gains (percent of the largest) and phases (degrees, with their signs) you type into the GUI in Steps 7 and 8. Fill in the table, and attach the script or its output.

Element	1	2	3	4	5	6	7	8
Gain (%)								
Phase (°)								

(c) Find $|r_{\mathrm{n}}|$ in dB. It is the uniform pattern's level at $+22.5^\circ$, the sidelobe level you read in Step 6. Then predict the main-lobe loss: the subtraction term $-20\log_{10}(1-|r_{\mathrm{n}}|^2)$, the rescaling term $20\log_{10}\max|w_n|$, and their total.

$|r_{\mathrm{n}}|$ =

loss =

(d) Two checks for Steps 7 to 9. Where does the notch land if every phase is entered with the wrong sign, and what do you see if you enter the gains but forget the phases?

2. How deep can the notch go? (Step 9, Table A). Lesson 24 estimated the residual of independent, one-step channel errors at the null as

$$\frac{|y(\theta_1)|}{N} \approx \frac{\epsilon_{\mathrm{rms}}}{\sqrt{N}}, \qquad \epsilon_{\mathrm{rms}} = \sqrt{\sigma_\phi^2 + \sigma_a^2}, \qquad \sigma_\phi = \frac{\mathrm{LSB}}{\sqrt{12}}.$$

Estimate the notch depth for the ADAR1000 (2.8125° phase steps, 1% gain steps), first relative to the null-steered peak and then relative to the uniform peak, which is how Table A reads it.

depth =

3. The difference null (Procedure B, Table B). Setting Beam 1 Phase to 180° subtracts the channel of elements 5–8 from the channel of elements 1–4:

$$\Delta(\theta) = \sum_{n=0}^{3} e^{jn\psi} - \sum_{n=4}^{7} e^{jn\psi}, \qquad \psi = kd\sin\theta.$$

(a) Show that Δ is zero at boresight at every frequency.

(b) Evaluate $|\Delta(\theta)|$ in Python over $-30^\circ \leqslant \theta \leqslant 30^\circ$ at your frequency, and find the angles of its two peaks.

peaks =

4. MVDR against an interferer (Procedure C, Table C). The simulated interferer arrives 10 dB stronger than the target at the aperture. The target sits at boresight, and the two digital channels sit four elements apart, so a source at θ presents a channel phase difference of $4kd \sin\theta$.

(a) In Manual, the beam is the uniform sum beam at broadside. Find its level at $+20^\circ$, and predict **Interferer vs Target**, the interferer's level in the output relative to the target's.

Int. vs Target =

(b) Find the channel phase difference for an interferer at $+20^\circ$ and at $+30^\circ$, between 0° and 360°. The target's is 0°.

at $+20^\circ$ =

at $+30^\circ$ =

(c) Use (b) to predict Step 21: with the interferer at $+30^\circ$, what happens to the target's Peak Gain under MVDR with almost no diagonal loading, and what does a loading of 0.1 change?

5. The monopulse pair (Procedure D, Table D). The Tracking preset forms the sum beam Σ and the difference beam Δ of Question 3 at once, and the kit plots the error $\varepsilon = \text{sign}\,(\angle\,\Sigma\Delta^*)\,|\Delta|/|\Sigma|$, positive when the source sits on the $+\theta$ side of the beam.

(a) With the HB100 at boresight, where should the delta null and the error's zero crossing sit in a beam sweep, and why?

null, zero =

(b) In that sweep, which side of the crossing reads positive: negative steer angles or positive ones? Explain from the sign convention.

(c) Near the beam $|\Delta|/|\Sigma| = |\tan 2\psi| \approx 2kd\,\theta$. Find the error's slope per degree at your frequency.

slope =

(d) How deep should the delta null read compared with your Table B difference null, and why?

Part 2: Measurements (your kit, in class)

Follow the procedure on the L24 Lab page, and record each reading as you take it.

Freq =	
Rx Gain =	
Distance =	

6. **Table A: the static notch** (Procedure A, Steps 6–10). Freeze the uniform trace first. If its two first sidelobes differ by more than about 2 dB, recalibrate before you go on. Read peaks with **Peak Gain** and angles with **Angle** set to **Peak**.

Quantity	**Measured**	**Predicted**
Uniform sidelobe nearest $+22.5^\circ$		
Null-steered main-lobe peak		
Notch bottom, below the uniform peak		
Notch angle		

7. **Table B: the difference null** (Procedure B, Steps 11–15). Beam 1 Phase at 180°, Mode Manual, uniform taper.

Quantity	**Measured**	**Predicted**
Null depth		
Null angle		
Left peak angle		
Right peak angle		

8. **Table C: MVDR against an interferer** (Procedure C, Steps 16–22). Simulated interferer at $+20^\circ$, Power 0 dBFS, about 10 dB above the target; target your HB100 at boresight; beam held at 0° in Signal vs Time; MVDR with $K = 1024$ snapshots. Each entry is the median of ten readings.

Mode	**Interferer vs Target**	**Peak Gain**	**MVDR Suppression**
Manual			
MVDR, load 0.1			

Adaptive tab	**Null angle**	**Null depth**	**Conventional at null**
MVDR, load 0.1			

Interferer at $+30^\circ$	**Manual**	**MVDR, load** 0.001	**MVDR, load** 0.1
Peak Gain			

9. **Table D: monopulse** (Procedure D, Steps 23–27). The Tracking preset, Manual beamformer, both digital beams at 0°, uniform taper.

Quantity	**Measured**	**Predicted**
Delta null depth		
Delta null angle		
Error zero crossing		
Error sign below, above		

Tracking	**How far it followed the moving HB100**
Step 27	

Stand's scale	-20°	-10°	$+10^\circ$	$+20^\circ$
Tracked angle				

Part 3: Writeup (individual, after the lab)

10. The notch you measured. Compare Table A with your Question 2 estimate. What set the depth of the notch on your kit? A short paragraph.

11. Reading a null-steered sweep. A student freezes the uniform trace, loads the null-steering table, and sweeps again. The null-steered peak reads -1.8 dB and the notch reads -48 dB, both relative to the uniform peak.

(a) Express the notch depth relative to the null-steered array's own peak, which is the number an interference calculation needs.

depth = ____

(b) A second student, with the same table, finds the level at $+22.5^\circ$ still near -13 dBc and a deep notch near -22.5° instead. Give the cause and what they should check.

(c) A third notch sits at $+25.7^\circ$, about 3° from the design angle. Which of these is large enough to explain it: rounding the weights, using the nominal 10.525 GHz instead of the kit's frequency, the sweep's step, or aligning the kit by eye?

12. MVDR on your kit. Use Table C and Question 4.

(a) Compare Manual's Interferer vs Target with your Question 4(a) prediction, and say how much MVDR took off the interferer at $+20^\circ$.

(b) Explain your $+30^\circ$ row with your Question 4(b) phase differences, and say why the lab runs MVDR at a loading of 0.1.

13. Choosing an approach. Each situation below uses the same 8-element PHASER.

(a) A fixed ground emitter sits at a surveyed bearing of $+35^\circ$ from the array face and does not move. The array must keep its beam on a target at -10°. Which approach would you use, and why?

(b) An airborne jammer of unknown and changing bearing appears during a mission. Which approach would you use, and why?

(c) A requirement calls for a 33 dB notch on a board whose 3-bit phase shifters allow a notch only about 22 dB deep (Lesson 24). Taking about 6 dB of improvement per added bit, how many phase bits are needed?

$B =$ ____

(d) Two jammers now illuminate the array from different bearings. State what happens when MVDR is asked to handle both on this hardware.

14. The monopulse pair. Two or three sentences each, using your Table D.

(a) Compare your tracked angles with the stand's scale. How closely do their sizes agree, and what explains the sign and any difference?

(b) Your delta null and your error's zero both sit a degree or two off 0°. What does that offset tell you about the kit, and does it matter to a tracker?

(c) The kit steps one LSB per update by the error's sign. What would stepping in proportion to the error, using your Question 5(c) slope, change?

Documentation: _____